Lab 7

Question 1

Practice 1

```
frank@cis106:~$ curl -s https://cis106.com/assets/scripts/wildcardpractice.sh | bash
--2026-04-13 20:08:49--  https://cis106.com/assets/logo100x100.png
Resolving cis106.com (cis106.com)... 185.199.108.153
Connecting to cis106.com (cis106.com)|185.199.108.153|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3520 (3.4K) [image/png]
Saving to: ‘logo100x100.png’

logo100x100.png                     100%[===================================================================================>]   3.44K  --.-KB/s    in 0s

2026-04-13 20:08:49 (105 MB/s) - ‘logo100x100.png’ saved [3520/3520]

frank@cis106:~$ ls wildcardpractice/*random*
wildcardpractice/_random.csv  wildcardpractice/random_text.md  wildcardpractice/randomtext.txt
frank@cis106:~$ ls wildcardpractice/*.cvs
ls: cannot access 'wildcardpractice/*.cvs': No such file or directory
frank@cis106:~$ ls wildcardpractice/*.txt
wildcardpractice/78food.txt  wildcardpractice/randomtext.txt
frank@cis106:~$ ls wildcardpractice/random*
wildcardpractice/random_text.md  wildcardpractice/randomtext.txt
frank@cis106:~$ ls wildcardpractice/random*.md
wildcardpractice/random_text.md
frank@cis106:~$
```

Practice 2

```
frank@cis106:~$ ls wildcardpractice/*.???
wildcardpractice/2020-08-27-capture.raw  wildcardpractice/baik.pdf   wildcardpractice/bk.pdf    wildcardpractice/file.log         wildcardpractice/sample1.doc
wildcardpractice/2022-02-02-camera.jpg   wildcardpractice/baik.xml   wildcardpractice/bk.xml    wildcardpractice/logo100x100.png  wildcardpractice/sample1.xls
wildcardpractice/78food.txt              wildcardpractice/beatk.exe  wildcardpractice/book.exe  wildcardpractice/_random.csv      wildcardpractice/sample.pdf
wildcardpractice/allfiles.lst            wildcardpractice/beatk.pdf  wildcardpractice/book.pdf  wildcardpractice/randomtext.txt   wildcardpractice/Sample.pdf
wildcardpractice/Allfiles.lst            wildcardpractice/beatk.xml  wildcardpractice/book.xml  wildcardpractice/Sample1234.csv
wildcardpractice/baik.exe                wildcardpractice/bk.exe     wildcardpractice/fail.log  wildcardpractice/sample123.csv
frank@cis106:~$ ls wildcardpractice/.??*
wildcardpractice/.bash_history  wildcardpractice/.hidden1.py   wildcardpractice/.hidden2.py
wildcardpractice/.hidden1.doc   wildcardpractice/.hidden2.doc  wildcardpractice/.hiddenFile.txt
frank@cis106:~$ ls wildcardpractice/b??*.???
wildcardpractice/baik.exe  wildcardpractice/baik.xml   wildcardpractice/beatk.pdf  wildcardpractice/book.exe  wildcardpractice/book.xml
wildcardpractice/baik.pdf  wildcardpractice/beatk.exe  wildcardpractice/beatk.xml  wildcardpractice/book.pdf
frank@cis106:~$ ls wildcardpractice/sample?.*
wildcardpractice/sample1.doc  wildcardpractice/sample1.xls
frank@cis106:~$ ls wildcardpractice/.??*.??
wildcardpractice/.hidden1.py  wildcardpractice/.hidden2.py
frank@cis106:~$
```

Practice 3

```
frank@cis106:~$ ls wildcardpractice/[a-z]*.???
wildcardpractice/allfiles.lst  wildcardpractice/beatk.pdf  wildcardpractice/book.exe  wildcardpractice/logo100x100.png  wildcardpractice/sample.pdf
wildcardpractice/baik.exe      wildcardpractice/beatk.xml  wildcardpractice/book.pdf  wildcardpractice/randomtext.txt
wildcardpractice/baik.pdf      wildcardpractice/bk.exe     wildcardpractice/book.xml  wildcardpractice/sample123.csv
wildcardpractice/baik.xml      wildcardpractice/bk.pdf     wildcardpractice/fail.log  wildcardpractice/sample1.doc
wildcardpractice/beatk.exe     wildcardpractice/bk.xml     wildcardpractice/file.log  wildcardpractice/sample1.xls
frank@cis106:~$ ls wildcardpractice/[A-Z]*[0-9]*
wildcardpractice/Sample1234.csv
frank@cis106:~$ ls wildcardpractice/[0-9][0-9][0-9][0-9]*
wildcardpractice/2020-08-27-capture.raw  wildcardpractice/2021-04-10-screenshot.jpeg  wildcardpractice/2022-02-02-camera.jpg
frank@cis106:~$ ls wildcardpractice/[!AEIOUaeiou]*
wildcardpractice/2020-08-27-capture.raw      wildcardpractice/beatk.pdf  wildcardpractice/fail.log         wildcardpractice/sample1.doc
wildcardpractice/2021-04-10-screenshot.jpeg  wildcardpractice/beatk.xml  wildcardpractice/file.log         wildcardpractice/sample1.xls
wildcardpractice/2022-02-02-camera.jpg       wildcardpractice/bk.exe     wildcardpractice/logo100x100.png  wildcardpractice/sample99.docx
wildcardpractice/78food.txt                  wildcardpractice/bk.pdf     wildcardpractice/_random.csv      wildcardpractice/samplejsonfile.json
wildcardpractice/baik.exe                    wildcardpractice/bk.xml     wildcardpractice/random_text.md   wildcardpractice/sample.pdf
wildcardpractice/baik.pdf                    wildcardpractice/book.exe   wildcardpractice/randomtext.txt   wildcardpractice/Sample.pdf
wildcardpractice/baik.xml                    wildcardpractice/book.pdf   wildcardpractice/Sample1234.csv   wildcardpractice/samplesite.html
wildcardpractice/beatk.exe                   wildcardpractice/book.xml   wildcardpractice/sample123.csv
frank@cis106:~$ ls wildcardpractice/[[:punct:]]
ls: cannot access 'wildcardpractice/[[:punct:]]': No such file or directory
frank@cis106:~$ ls wildcardpractice/[[:punct:]]*
wildcardpractice/_random.csv
frank@cis106:~$
```

Practice 4

```
frank@cis106:~$ mkdir -p music/{jazz,rock}/{mp3files,videos,oggfiles}/new
frank@cis106:~$ tree music/
music/
├── jazz
│   ├── mp3files
│   │   └── new
│   ├── oggfiles
│   │   └── new
│   └── videos
│       └── new
└── rock
    ├── mp3files
    │   └── new
    ├── oggfiles
    │   └── new
    └── videos
        └── new

15 directories, 0 files
frank@cis106:~$
```

Question 2

Directory 1

```
frank@cis106:~$ mkdir -p wallpaper/cars/{1080p,2k,4k}
frank@cis106:~$ tree wallpaper
wallpaper
└── cars
    ├── 1080p
    ├── 2k
    └── 4k

5 directories, 0 files
frank@cis106:~$
```

Directory 2

```
frank@cis106:~$ mkdir -p assets/{imgs,video}/{large,small}
frank@cis106:~$ ls assets/
imgs  video
frank@cis106:~$ tree assets/
assets/
├── imgs
│   ├── large
│   └── small
└── video
    ├── large
    └── small

7 directories, 0 files
frank@cis106:~$
```

Question 3

```
Tilix: frank@cis106: ~/website_ch1_lab7
1: frank@cis106: ~/website_ch1_lab7
frank@cis106:~$ mkdir -p website_ch1_lab7/{app,assets{css,images},components,config,data,pages}
frank@cis106:~$ tree website_ch1_lab7/
website_ch1_lab7/
├── 404.html
├── api.js
├── app
├── APP.conf
├── APP.conf.bk
├── app.js
├── assetscss
├── assetsimages
├── backup.sh
├── build.sh
├── cart.svg
├── components
├── config
├── data
├── deploy.sh
├── footer.html
├── footer.jpg
├── header.html
├── hero.png
├── index.html
├── item1.raw
├── item2.jpg
├── navbar.html
├── pages
├── post1.html
├── post2.html
├── post3.html
├── posts.json
├── _Products.Json
├── _reset.css
├── Robots.txt
├── sidebar.html
├── _style.css
├── _theme.css
```

```
Tilix: frank@cis106: ~/website_ch1_lab7
1: frank@cis106: ~/website_ch1_lab7
8 directories, 29 files
frank@cis106:~$ tree website_ch1_lab7/assets
website_ch1_lab7/assets  [error opening dir]

0 directories, 0 files
frank@cis106:~$ rmdir website_ch1_lab7/assetscss/
frank@cis106:~$ rmdir website_ch1_lab7/assetsimages/
frank@cis106:~$ tree website_ch1_lab7/
website_ch1_lab7/
├── 404.html
├── api.js
├── app
├── APP.conf
├── APP.conf.bk
├── app.js
├── backup.sh
├── build.sh
├── cart.svg
├── components
├── config
├── data
├── deploy.sh
├── footer.html
├── footer.jpg
├── header.html
├── hero.png
├── index.html
├── item1.raw
├── item2.jpg
├── navbar.html
├── pages
├── post1.html
├── post2.html
├── post3.html
├── posts.json
├── _Products.Json
├── _reset.css
```

```
6 directories, 29 files
frank@cis106:~$ mkdir -p website_ch1_lab7/assets/{css,images}
frank@cis106:~$ tree website_ch1_lab7/
website_ch1_lab7/
├── 404.html
├── api.js
├── app
├── APP.conf
├── APP.conf.bk
├── app.js
├── assets
│   ├── css
│   └── images
├── backup.sh
├── build.sh
├── cart.svg
├── components
├── config
├── data
├── deploy.sh
├── footer.html
├── footer.jpg
├── header.html
├── hero.png
├── index.html
├── item1.raw
├── item2.jpg
├── navbar.html
├── pages
├── post1.html
├── post2.html
├── post3.html
├── posts.json
├── _Products.Json
├── _reset.css
├── Robots.txt
├── sidebar.html
```

```
└── Users.JSON

9 directories, 29 files
frank@cis106:~$ ls website_ch1_lab7/*.css
website_ch1_lab7/_reset.css  website_ch1_lab7/_style.css  website_ch1_lab7/_theme.css
frank@cis106:~$ mv website_ch1_lab7/*.css website_ch1_lab7/assets/css/
frank@cis106:~$ tree website_ch1_lab7/
website_ch1_lab7/
├── 404.html
├── api.js
├── app
├── APP.conf
├── APP.conf.bk
├── app.js
├── assets
│   ├── css
│   │   ├── _reset.css
│   │   ├── _style.css
│   │   └── _theme.css
│   └── images
├── backup.sh
├── build.sh
├── cart.svg
├── components
├── config
├── data
├── deploy.sh
├── footer.html
├── footer.jpg
├── header.html
├── hero.png
├── index.html
├── item1.raw
├── item2.jpg
├── navbar.html
├── pages
├── post1.html
```

```
Tilix: frank@cis106: ~/website_ch1_lab7
1: frank@cis106: ~/website_ch1_lab7
└── Users.JSON

9 directories, 29 files
frank@cis106:~$ ls website_ch1_lab7/post*.html
website_ch1_lab7/post1.html  website_ch1_lab7/post2.html  website_ch1_lab7/post3.html
frank@cis106:~$ mv website_ch1_lab7/post*.html website_ch1_lab7/pages/
frank@cis106:~$ tree website_
website_  [error opening dir]

0 directories, 0 files
frank@cis106:~$ tree website_ch1_lab7/
website_ch1_lab7/
├── 404.html
├── api.js
├── app
├── APP.conf
├── APP.conf.bk
├── app.js
├── assets
│   ├── css
│   │   ├── _reset.css
│   │   ├── _style.css
│   │   └── _theme.css
│   └── images
├── backup.sh
├── build.sh
├── cart.svg
├── components
├── config
├── data
├── deploy.sh
├── footer.html
├── footer.jpg
├── header.html
├── hero.png
├── index.html
├── item1.raw
```

```
Tilix: frank@cis106: ~/website_ch1_lab7
1: frank@cis106: ~/website_ch1_lab7
└── Users.JSON

9 directories, 29 files
frank@cis106:~$ ls website_ch1_lab7/*.json
website_ch1_lab7/posts.json
frank@cis106:~$ ls website_ch1_lab7/*.????
website_ch1_lab7/404.html     website_ch1_lab7/navbar.html
website_ch1_lab7/APP.conf     website_ch1_lab7/posts.json
website_ch1_lab7/footer.html  website_ch1_lab7/_Products.Json
website_ch1_lab7/header.html  website_ch1_lab7/sidebar.html
website_ch1_lab7/index.html   website_ch1_lab7/Users.JSON
frank@cis106:~$ ls website_ch1_lab7/[a-z]*.html
website_ch1_lab7/footer.html  website_ch1_lab7/navbar.html
website_ch1_lab7/header.html  website_ch1_lab7/sidebar.html
website_ch1_lab7/index.html
frank@cis106:~$ ls website_ch1_lab7/*r.html
website_ch1_lab7/footer.html  website_ch1_lab7/navbar.html
website_ch1_lab7/header.html  website_ch1_lab7/sidebar.html
frank@cis106:~$ mv website_ch1_lab7/*r.html website_ch1_lab7/components/
frank@cis106:~$ tree website_
website_  [error opening dir]

0 directories, 0 files
frank@cis106:~$ tree website_ch1_lab7/
website_ch1_lab7/
├── 404.html
├── api.js
├── app
├── APP.conf
├── APP.conf.bk
├── app.js
├── assets
│   ├── css
│   │   ├── _reset.css
│   │   ├── _style.css
│   │   └── _theme.css
│   └── images
├── backup.sh
```

```
Tilix: frank@cis106: ~/website_ch1_lab7
1: frank@cis106: ~/website_ch1_lab7
├── Robots.txt
└── Users.JSON

9 directories, 29 files
frank@cis106:~$ ls website_ch1_lab7/*.conf.*
website_ch1_lab7/APP.conf.bk
frank@cis106:~$ cd website_ch1_lab7/
frank@cis106:~/website_ch1_lab7$ ls *.conf.*
APP.conf.bk
frank@cis106:~/website_ch1_lab7$ ls *.conf*
APP.conf  APP.conf.bk
frank@cis106:~/website_ch1_lab7$ ls *conf*
APP.conf  APP.conf.bk

config:
frank@cis106:~/website_ch1_lab7$ ls *con*
APP.conf  APP.conf.bk

config:
frank@cis106:~/website_ch1_lab7$ mv *conf* config/
mv: cannot move 'config' to a subdirectory of itself, 'config/config'
frank@cis106:~/website_ch1_lab7$ mv *conf.* config/
mv: cannot stat '*conf.*': No such file or directory
frank@cis106:~/website_ch1_lab7$ tree
.
├── 404.html
├── api.js
├── app
├── app.js
├── assets
│   ├── css
│   │   ├── _reset.css
│   │   ├── _style.css
│   │   └── _theme.css
│   └── images
├── backup.sh
├── build.sh
├── cart.svg
```

```
Tilix: frank@cis106: ~/website_ch1_lab7
1: frank@cis106: ~/website_ch1_lab7
├── Robots.txt
└── Users.JSON

9 directories, 29 files
frank@cis106:~/website_ch1_lab7$ mkdir config/deploy
frank@cis106:~/website_ch1_lab7$ tree
.
├── 404.html
├── api.js
├── app
├── app.js
├── assets
│   ├── css
│   │   ├── _reset.css
│   │   ├── _style.css
│   │   └── _theme.css
│   └── images
├── backup.sh
├── build.sh
├── cart.svg
├── components
│   ├── footer.html
│   ├── header.html
│   ├── navbar.html
│   └── sidebar.html
├── config
│   ├── APP.conf
│   ├── APP.conf.bk
│   ├── DB1.conf.old
│   └── deploy
├── data
├── deploy.sh
├── footer.jpg
├── hero.png
├── index.html
├── item1.raw
├── item2.jpg
├── pages
```

```
└── Users.JSON

10 directories, 29 files
frank@cis106:~/website_ch1_lab7$ ls *.sh
backup.sh  build.sh  deploy.sh
frank@cis106:~/website_ch1_lab7$ mv *.sh config/deploy/
frank@cis106:~/website_ch1_lab7$ tree
.
├── 404.html
├── api.js
├── app
├── app.js
├── assets
│   ├── css
│   │   ├── _reset.css
│   │   ├── _style.css
│   │   └── _theme.css
│   └── images
├── cart.svg
├── components
│   ├── footer.html
│   ├── header.html
│   ├── navbar.html
│   └── sidebar.html
├── config
│   ├── APP.conf
│   ├── APP.conf.bk
│   ├── 401.conf.old
│   └── deploy
│       ├── backup.sh
│       ├── build.sh
│       └── deploy.sh
├── data
├── footer.jpg
├── hero.png
├── index.html
├── item1.raw
├── item2.jpg
```

```
├── Robots.txt
└── Users.JSON

10 directories, 29 files
frank@cis106:~/website_ch1_lab7$ ls *s.????
posts.json  _Products.Json  Users.JSON
frank@cis106:~/website_ch1_lab7$ mv *s.???? data/
frank@cis106:~/website_ch1_lab7$ tree
.
├── 404.html
├── api.js
├── app
├── app.js
├── assets
│   ├── css
│   │   ├── _reset.css
│   │   ├── _style.css
│   │   └── _theme.css
│   └── images
├── cart.svg
├── components
│   ├── footer.html
│   ├── header.html
│   ├── navbar.html
│   └── sidebar.html
├── config
│   ├── APP.conf
│   ├── APP.conf.bk
│   ├── 401.conf.old
│   └── deploy
│       ├── backup.sh
│       ├── build.sh
│       └── deploy.sh
├── data
│   ├── posts.json
│   ├── _Products.Json
│   └── Users.JSON
├── footer.jpg
```

```
Tilix: frank@cis106: ~/website_ch1_lab7
1: frank@cis106: ~/website_ch1_lab7
│   ├── post2.html
│   └── post3.html
└── Robots.txt

10 directories, 29 files
frank@cis106:~/website_ch1_lab7$ ls ap*
api.js  app.js

app:
frank@cis106:~/website_ch1_lab7$ mv ap* ap
api.js  app/    app.js
frank@cis106:~/website_ch1_lab7$ tree
.
├── 404.html
├── api.js
├── app
├── app.js
├── assets
│   ├── css
│   │   ├── _reset.css
│   │   ├── _style.css
│   │   └── _theme.css
│   └── images
├── cart.svg
├── components
│   ├── footer.html
│   ├── header.html
│   ├── navbar.html
│   └── sidebar.html
├── config
│   ├── APP.conf
│   ├── APP.conf.bk
│   ├── 401.conf.old
│   └── deploy
│       ├── backup.sh
│       ├── build.sh
│       └── deploy.sh
├── data
```

```
Tilix: frank@cis106: ~/website_ch1_lab7
1: frank@cis106: ~/website_ch1_lab7
│   ├── post2.html
│   └── post3.html
└── Robots.txt

10 directories, 29 files
frank@cis106:~/website_ch1_lab7$ ls *ap
ls: cannot access '*ap': No such file or directory
frank@cis106:~/website_ch1_lab7$ tree
.
├── 404.html
├── api.js
├── app
├── app.js
├── assets
│   ├── css
│   │   ├── _reset.css
│   │   ├── _style.css
│   │   └── _theme.css
│   └── images
├── cart.svg
├── components
│   ├── footer.html
│   ├── header.html
│   ├── navbar.html
│   └── sidebar.html
├── config
│   ├── APP.conf
│   ├── APP.conf.bk
│   ├── 401.conf.old
│   └── deploy
│       ├── backup.sh
│       ├── build.sh
│       └── deploy.sh
├── data
│   ├── posts.json
│   ├── _Products.Json
│   └── Users.JSON
├── footer.jpg
```

```
Tilix: frank@cis106: ~/website_ch1_lab7
1: frank@cis106: ~/website_ch1_lab7
│   ├── post2.html
│   └── post3.html
└── Robots.txt

10 directories, 29 files
frank@cis106:~/website_ch1_lab7$ ls *.js
api.js  app.js
frank@cis106:~/website_ch1_lab7$ mv *.js app/
frank@cis106:~/website_ch1_lab7$ tree
.
├── 404.html
├── app
│   ├── api.js
│   └── app.js
├── assets
│   ├── css
│   │   ├── _reset.css
│   │   ├── _style.css
│   │   └── _theme.css
│   └── images
├── cart.svg
├── components
│   ├── footer.html
│   ├── header.html
│   ├── navbar.html
│   └── sidebar.html
├── config
│   ├── APP.conf
│   ├── APP.conf.bk
│   ├── [illegible]
│   └── deploy
│       ├── backup.sh
│       ├── build.sh
│       └── deploy.sh
├── data
│   ├── posts.json
│   ├── _Products.Json
│   └── Users.JSON
```

```
Tilix: frank@cis106: ~/website_ch1_lab7
1: frank@cis106: ~/website_ch1_lab7

10 directories, 29 files
frank@cis106:~/website_ch1_lab7$ mv cart.svg footer.jpg hero.png item1.raw item2.jpg images/
mv: target 'images/': No such file or directory
frank@cis106:~/website_ch1_lab7$ mv cart.svg footer.jpg hero.png item1.raw item2.jpg assets/images/
frank@cis106:~/website_ch1_lab7$ tree
.
├── 404.html
├── app
│   ├── api.js
│   └── app.js
├── assets
│   ├── css
│   │   ├── _reset.css
│   │   ├── _style.css
│   │   └── _theme.css
│   └── images
│       ├── cart.svg
│       ├── footer.jpg
│       ├── hero.png
│       ├── item1.raw
│       └── item2.jpg
├── components
│   ├── footer.html
│   ├── header.html
│   ├── navbar.html
│   └── sidebar.html
├── config
│   ├── APP.conf
│   ├── APP.conf.bk
│   ├── [illegible]
│   └── deploy
│       ├── backup.sh
│       ├── build.sh
│       └── deploy.sh
├── data
│   ├── posts.json
│   ├── _Products.Json
│   └── Users.JSON
├── index.html
├── pages
│   ├── post1.html
│   ├── post2.html
│   └── post3.html
└── Robots.txt

10 directories, 29 files
frank@cis106:~/website_ch1_lab7$
```